1. k-NN Classification with Varying k

Dataset: Iris Dataset

Tasks

i. Load the dataset and split it into training and testing sets (80:20).

ii. Train a k-NN classifier for k = 1, 3, 5, 7 and 9.

iii. Compute the accuracy for each value of k.

iv. Plot accuracy versus k.

v. Identify the best value of k.

2. Effect of Feature Scaling on k-NN

Dataset: Wine Dataset

Tasks

i. Load the dataset and split it into training and testing sets.

ii. Train k-NN (k = 5) without scaling and compute accuracy.

iii. Apply StandardScaler and train k-NN again.

iv. Compare both accuracies.

v. Explain why scaling is important for distance-based algorithms.

3. Wine Classification using Gaussian Naïve Bayes

Dataset: Wine Dataset

Tasks

i. Load the dataset and perform a train-test split.

ii. Train a GaussianNB classifier.

iii. Predict the test samples.

iv. Print the classification report.

v. Plot the confusion matrix as a heatmap.

vi. Identify the most frequently misclassified class.

4. Decision Tree using ID3 and CART Criteria

Dataset: Wine Dataset

Tasks

i. Load the dataset and split it into training and testing sets.

ii. Train a Decision Tree with criterion="entropy" (ID3-style).

iii. Train a Decision Tree with criterion="gini" (CART).

iv. Compute Accuracy, Precision, Recall and F1-score for both.

v. Compare the results and state which performs better.

5. Decision Tree Depth and Overfitting

Dataset: Wine Dataset

Tasks

i. Load the dataset and split it into training and testing sets.

ii. Train Decision Trees with max_depth from 1 to 10.

iii. Record training and testing accuracy for each depth.

iv. Plot both accuracy curves.

v. Identify the depth at which overfitting begins.

6. Decision Tree Evaluation using 5-Fold Cross-Validation

Dataset: Wine Dataset

Tasks

i. Load the dataset.

ii. Train a Decision Tree classifier.
iii. Perform 5-Fold Cross-Validation.
iv. Display the fold-wise accuracy.
v. Calculate the mean accuracy and standard deviation.

7. Ice-Cream Sales Prediction using Simple Linear Regression
Dataset: Ice Cream Sales Dataset
Tasks
i. Load the dataset and select temperature as the input and sales as the target.
ii. Split the dataset into training and testing sets.
iii. Train a Simple Linear Regression model.
iv. Compute MAE, MSE, RMSE and R^2 score.
v. Plot temperature versus actual and predicted sales.
vi. Predict the expected sales at 35 °C.

8. House Price Prediction using Multiple Linear Regression
Dataset: California Housing Dataset
Tasks
i. Load the dataset and identify the independent and dependent variables.
ii. Split the dataset into training and testing sets.
iii. Train a Multiple Linear Regression model.
iv. Compute MAE, MSE, RMSE and R^2 score.
v. Plot actual versus predicted house prices.
vi. Print the coefficients and identify the most influential feature.

9. Classification using Logistic Regression
Dataset: Iris Dataset
Tasks
i. Load the dataset and split it into training and testing sets.
ii. Train a Logistic Regression classifier.
iii. Predict the class labels for the test samples.
iv. Compute accuracy, display the confusion matrix and print the classification report.
v. Print the predicted class probabilities for any three test samples.

10. Iris Flower Classification using Linear SVM
Dataset: Iris Dataset
Tasks
i. Load the dataset and separate the input features and target variable.
ii. Split the dataset into training and testing sets.
iii. Train a Linear SVM classifier.
iv. Compute Accuracy, Precision, Recall and F1-score.
v. Display the confusion matrix.
vi. Predict the species for the measurements [5.1, 3.5, 1.4, 0.2].

11. Non-Linear SVM using Different Kernels
Dataset: Wine Dataset and make_moons synthetic data
Tasks
i. Train SVM classifiers with Linear, Polynomial and RBF kernels on the Wine dataset.
ii. Tabulate the accuracy of each kernel.
iii. Generate make_moons data and train the same three kernels.

iv. Plot the decision boundaries.
v. State which kernel separates the non-linear data best and why.

12. K-Means Clustering with the Elbow Method
Dataset: Synthetic data from make_blobs
Tasks
i. Generate the dataset and visualize it.
ii. Compute WCSS for K = 1 to 10 and plot the elbow curve.
iii. Choose an appropriate value of K.
iv. Train the K-Means model and assign cluster labels.
v. Plot the clusters and centroids.
vi. Calculate the Silhouette Score.

13. EM Algorithm (Gaussian Mixture) versus K-Means
Dataset: Synthetic data from make_blobs (or Iris)
Tasks
i. Load or generate the dataset.
ii. Apply K-Means clustering.
iii. Apply GaussianMixture (EM algorithm) with the same number of clusters.
iv. Plot the clusters from both methods.
v. Compare the Silhouette Scores.
vi. Print the soft-assignment probabilities for three samples and explain the difference between hard and soft clustering.

14. Classification using Random Forest
Dataset: Iris Dataset
Tasks
i. Load the dataset and split it into training and testing sets.
ii. Train a Random Forest classifier with 100 trees.
iii. Predict the class labels for the test samples.
iv. Compute Accuracy, Precision, Recall and F1-score.
v. Display the confusion matrix.
vi. Plot the feature importances.

15. Boosting Algorithms: AdaBoost and XGBoost
Dataset: Wine Dataset
Tasks
i. Load the dataset and split it into training and testing sets.
ii. Train an AdaBoost classifier.
iii. Train an XGBoost classifier.
iv. Train a Random Forest classifier for comparison.
v. Tabulate the accuracy of all three models.
vi. Explain the difference between bagging and boosting.

16. Iris Classification using Single-Layer Perceptron
Dataset: Iris Dataset (Setosa and Versicolor only)
Tasks
i. Load the dataset and select two classes and two features.
ii. Split the dataset into training and testing sets.
iii. Train a Perceptron classifier.

iv. Compute Accuracy, Precision, Recall and F1-score.
v. Display the confusion matrix.
vi. Plot the decision boundary.

17. XOR Gate Classification using Multi-Layer Perceptron

Dataset: XOR Gate (Synthetic Input)

Tasks

i. Create the input-output dataset for the XOR gate.
ii. Train an MLPClassifier with one hidden layer.
iii. Predict the outputs for all input combinations.
iv. Display the predicted and actual outputs.
v. Calculate the classification accuracy.
vi. Train a single-layer perceptron on the same data and explain why it fails.